

Research Notes on Chain of thought reasoning

Chain of thought reasoning

>> Section 1

Multi-shot A prompt may include a few examples for a model to learn from in context, an approach called few-shot learning. For example, the prompt may ask the model to complete "maison → house, chat → cat, chien →", with the expected response being dog.

>> Section 2

Chain-of-thought (CoT) prompting is a technique that allows large language models (LLMs) to solve a problem as a series of intermediate steps before giving a final answer. In 2022, Google Brain reported that chain-of-thought prompting improves reasoning ability by inducing the model to answer a multi-step problem with steps of reasoning that mimic a train of thought. Chain-of-thought techniques were developed to help LLMs handle multi-step reasoning tasks, such as arithmetic or commonsense reasoning questions. When applied to PaLM, a 540 billion parameter language model, according to Google, CoT prompting significantly aided the model, allowing it to perform comparably with task-specific fine-tuned models on several tasks, achieving state-of-the-art results at the time on the GSM8K mathematical reasoning benchmark. It is possible to fine-tune models on CoT reasoning datasets to enhance this capability further and stimulate better interpretability. As originally proposed by Google, each CoT prompt is accompanied by a set of input/output examples—called exemplars—to demonstrate the desired model output, making it a few-shot prompting technique. However, according to a later paper from researchers at Google and the University of Tokyo, simply appending the words "Let's think step-by-step" was also effective, which allowed for CoT to be employed as a zero-shot technique.

>> Section 3

Textual inversion and embeddings For text-to-image models, textual inversion performs an optimization process to create a new word embedding based on a set of example images. This embedding vector acts as a "pseudo-word" which can be included in a prompt to express the content or style of the examples.

>> Section 4

History Early precedents of structured user interaction with ruled-based AI systems can be found in enterprise automation software from 1990s. For example, The Intelligent Filing Manager (1999), developed by Krishna C. Mukherjee, used a dynamic Q&A interface driven by rule-based expert system to collect user inputs for generating regulatory filings automatically across jurisdictions. While not involving neural networks, such

systems featured prompt-like workflows that influenced later human-in-the-loop AI designs. In 2018, researchers first proposed that all previously separate tasks in natural language processing (NLP) could be cast as question-answer problems over a context. In addition, they trained a first single, joint, multi-task model that would answer any task-related question like "What is the sentiment" or "Translate this sentence to German" or "Who is the president?" The AI boom saw an increased focus within academic literature and professional practice on applying prompting techniques to get the model to output the desired outcome and avoid nonsensical output, a process characterized by trial-and-error. After the release of ChatGPT in 2022, prompt engineering was soon seen as an important business skill; companies began hiring dedicated prompt engineers, although, given advances in AI's ability to generate prompts better than humans, the employment market for prompt engineers has faced uncertainty. According to The Wall Street Journal in 2025, the job of prompt engineer was one of the hottest in 2023, but has become obsolete due to models that better intuit user intent and to company trainings. A repository for prompts reported that over 2,000 public prompts for around 170 datasets were available in February 2022. In 2022, the chain-of-thought prompting technique was proposed by Google researchers. In 2023, several text-to-text and text-to-image prompt databases were made publicly available. The Personalized Image-Prompt (PIP) dataset, a generated image-text dataset that has been categorized by 3,115 users, has also been made available publicly in 2024.

>> Section 5

GraphRAG (coined by Microsoft Research) is a technique that extends RAG with the use of a knowledge graph to allow the model to connect disparate pieces of information, synthesize insights, and understand summarized semantic concepts over large data collections. It was shown to be effective on datasets like the Violent Incident Information from News Articles.

>> Section 6

Terminology The Oxford English Dictionary defines prompt engineering as "The action or process of formulating and refining prompts for an artificial intelligence program, algorithm, etc., in order to optimize its output or to achieve a desired outcome; the discipline or profession concerned with this." In 2023, prompt ("an instruction given to an artificial intelligence program, algorithm, etc., which determines or influences the content it generates") was the runner-up to Oxford's word of the year.

>> Section 7

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MIPRO (Multi-prompt Instruction Proposal Optimizer) optimizes the instructions and few-shot demonstrations of multi-stage language model programs, proposing small changes to module prompts and retaining those that improve a downstream performance metric without access to module-level labels or gradients. GEPA (Genetic-Pareto) is a reflective prompt optimizer for compound AI systems that combines language-model-based analysis of execution traces and textual feedback with a Pareto-based evolutionary search over a population of candidate systems; across four tasks, GEPA reports average gains of about 10% over reinforcement-learning-based Group Relative Policy Optimization (GRPO) and over 10% over the MIPROv2 prompt optimizer, while using up to 35 times fewer rollouts than GRPO. Open-source frameworks such as DSPy and Opik expose these and related optimizers, allowing prompt search to be expressed as part of a programmatic pipeline rather than through manual trial and error.

>> Section 8

Using gradient descent to search for prompts In "prefix-tuning", "prompt tuning", or "soft prompting", floating-point vectors are searched directly by gradient descent to maximize the log-likelihood on outputs. An earlier result uses the same idea of gradient descent search, but is designed for masked language models like BERT, and searches only over token sequences, rather than numerical vectors. Formally, it searches for $\arg \max_{\tilde{X}} \sum_i \log \Pr[Y_i | X \sim * \tilde{X}_i]$ where $X \sim$ ranges over token sequences of a specified length.

>> Section 9

Automated prompt generation Recent research has explored automated prompt engineering, using optimization algorithms to generate or refine prompts without human intervention. These automated approaches aim to identify effective prompt patterns by analyzing model gradients, reinforcement feedback, or evolutionary processes, reducing the need for manual experimentation.